

A close-up, slightly blurred image of a laptop screen displaying a candlestick chart for Apple (AAPL) stock. The chart shows price movement over time, with green candles indicating upward price movement and red candles indicating downward movement. A white trend line is drawn across the chart, showing an overall upward trend. The background is dark, and the laptop keyboard is visible in the bottom right corner.

AAPL MARKET

What's the best strategy to trade in AAPL

DATASET INTRODUCTION

The dataset has 5,124 rows representing individual trading days over approximately 20 years, with 7 columns: Date, Open, High, Low, Close, Adjusted Close, and Volume

DATASET OVERVIEW:

SHAPE: 5124 ROWS × 7 COLUMNS

DATE RANGE: 1999-11-18 TO 2020-04-01

TRADING DAYS: 5124

APPROXIMATE YEARS: 20.3

INITIAL DATA EXPLORATION

 Date	...	# Open	# High	# Low	# Close	# Adj Close	# Volume
2010-01-06		22.06723976135254	22.174535751342773	22.00286102294922	22.06723976135254	20.142688751220703	3243700
2012-06-19		28.9484977722168	29.556509017944336	28.89127349853516	29.434907913208008	26.95255279541016	4213800
2006-03-09		26.2804012298584	26.316165924072266	25.8726749420166	25.901287078857425	22.27627944946289	4092700
2003-09-02		17.45350456237793	17.69670867919922	17.310443878173828	17.646638870239258	15.176909446716309	4944500
2014-09-15		41.84549331665039	41.84549331665039	41.30186080932617	41.43061447143555	39.157859802246094	2266200
2009-04-20		12.131616592407228	12.16738224029541	11.552217483520508	11.74535083770752	10.721004486083984	6736600
2013-02-05		31.87410545349121	32.23175811767578	31.723892211914062	32.1888427734375	29.793331146240234	3020300
2008-09-19		22.896995544433597	23.60515022277832	22.7038631439209	23.540773391723636	21.487712860107425	4986200
2013-03-08		31.11587905883789	31.130186080932617	30.77253150939941	30.779685974121094	28.489049911499023	4552300
2000-09-29		34.55829620361328	35.586551666259766	34.200645446777344	35.00536346435547	30.1061954498291	3677400

	Open	High	Low	Close	Adj Close	\	Volume
count	5124.000000	5124.000000	5124.000000	5124.000000	5124.000000	count	5.124000e+03
mean	34.090255	34.560553	33.629467	34.106245	31.778674	mean	3.693250e+06
std	18.608831	18.834528	18.381718	18.611595	18.730529	std	2.481855e+06
min	7.653791	7.961373	7.510730	7.761087	6.674886	min	2.719000e+05
25%	21.101574	21.452074	20.785050	21.130186	18.790381	25%	2.206475e+06
50%	27.328326	27.703863	27.010015	27.396280	24.714866	50%	3.174050e+06
75%	41.500000	41.860001	41.130001	41.525204	39.203083	75%	4.508075e+06
max	111.587982	115.879829	103.719597	113.733902	97.816307	max	6.254630e+07

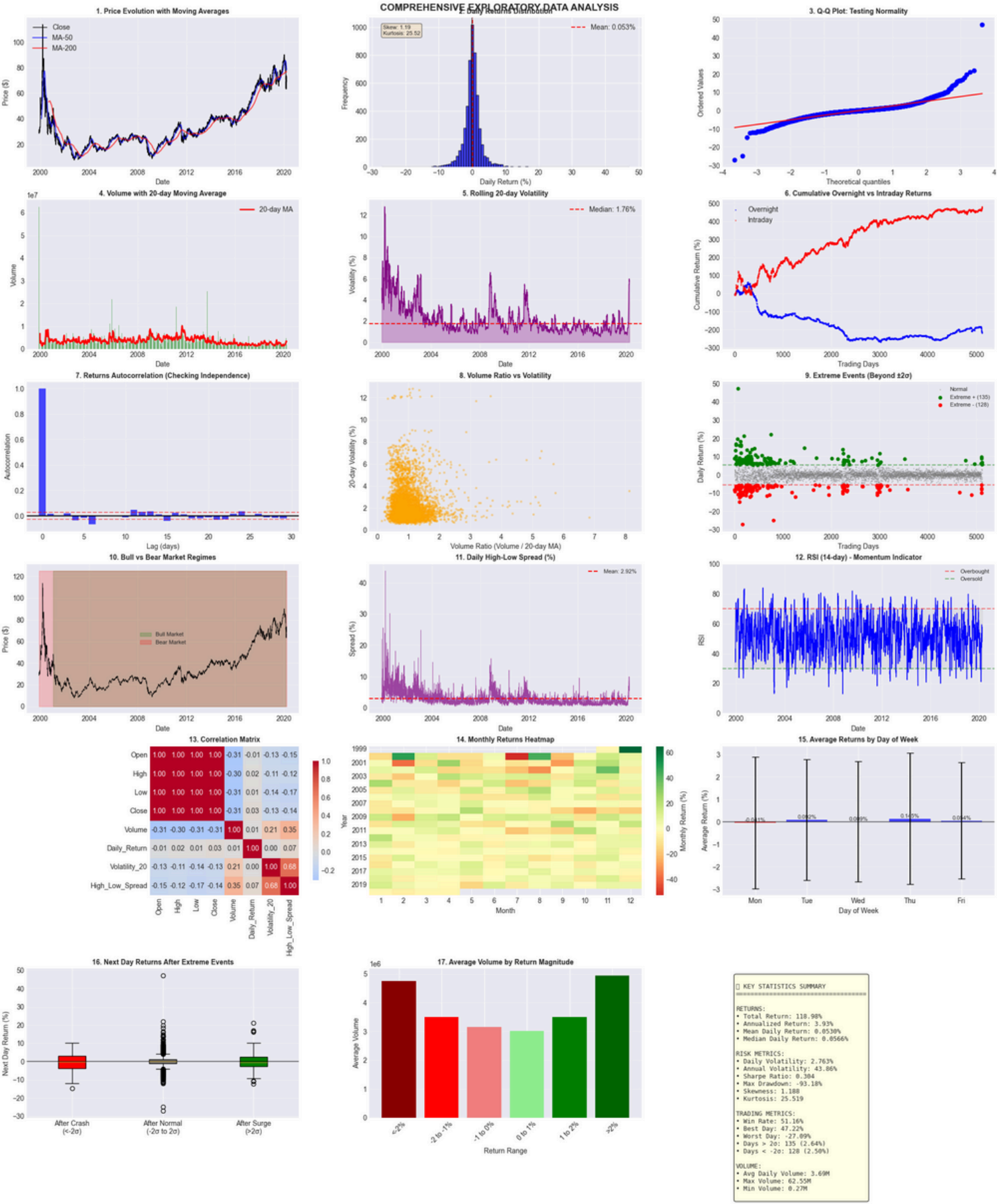
PRICE STATISTICS:
CLOSE PRICE – MEAN: \$34.11
CLOSE PRICE – MIN: \$7.76
CLOSE PRICE – MAX: \$113.73
TOTAL RETURN: 118.98%

FEATURE ENGINEERING

AT FIRST GLANCE, EVERYTHING LOOKS NORMAL – NO MISSING DATA, NO OBVIOUS ERRORS. BUT RAW PRICES DON'T TELL US THE FULL STORY.

1. `DAILY_RETURN (PCT_CHANGE() * 100)`
2. `LOG_RETURN`
3. `OVERNIGHT_RETURN (OPEN / PREVIOUS CLOSE - 1)`
4. `INTRADAY_RETURN (CLOSE / OPEN - 1)`
5. `HIGH_LOW_SPREAD`
6. `VOLUME_MA_20` AND `VOLUME_RATIO`
7. `MA_50` AND `MA_200`
8. `VOLATILITY_20`
9. `RSI`

LOOKING FOR REDFLAG IN THE DATA



1. RETURN DISTRIBUTION ANOMALIES:

- SKEWNESS: 1.188 (POSITIVE SKEW – MORE SURGES)
- EXCESS KURTOSIS: 25.519 (FAT TAILS – EXTREME EVENTS)
- MEAN RETURN: 0.0530% (POSITIVE DRIFT)
- (H_0): AVERAGE DAILY RETURN = 0 (NO DAILY PROFIT/LOSS TREND).

2. VOLATILITY CLUSTERING:

- VOLATILITY AUTOCORRELATION: 0.988
- STRONG CLUSTERING
- H_0 : AVERAGE RETURN DURING HIGH VOLATILITY = DURING LOW VOLATILITY.

3. VOLUME IMPACT:

- FLAG RULE (HIGH VOLUME): PRE-MARKET VOLUME > 20-DAY MA (VOLUME RATIO > 1)
- MEDIAN 20-DAY VOL ON FLAGGED DAYS: $\approx 3.24\%$
- MEDIAN 20-DAY VOL ON NORMAL DAYS: $\approx 1.76\%$
- VOLATILITY UPLIFT RATIO: $\approx 1.84\times$
- EXTREME-DAY FREQUENCY ($|\text{RETURN}| > 2\sigma$): $\approx 2.0\times$ HIGHER ON FLAGGED DAYS
- Z-STATISTIC (DIFFERENCE TEST): 15.17 ($P \approx 0.0000$)
- H_0 : HIGH-VOLUME DAYS ARE NO DIFFERENT FROM NORMAL DAYS IN TERMS OF MAGNITUDE OF MOVEMENT (VOLATILITY/EXTREME EVENTS).

4. DAY-OF-WEEK ANOMALY:

- MONDAY AVERAGE: -0.0407%
- OTHER DAYS AVERAGE: 0.0749%
- MONDAY EFFECT: -0.1156%
- H_0 : AVERAGE RETURN ON MONDAY = OTHER DAYS.

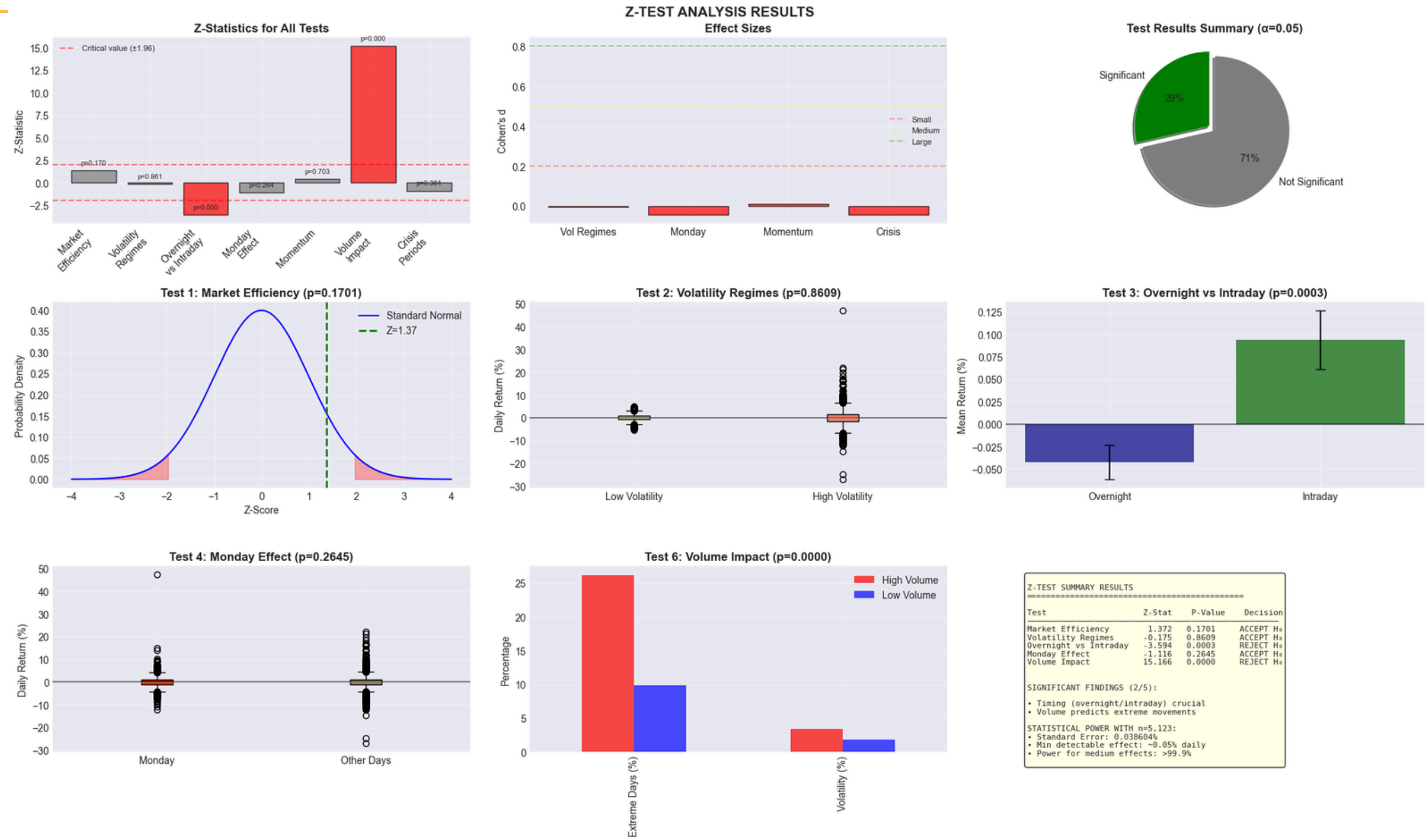
5. OVERNIGHT ANOMALY:

- TOTAL OVERNIGHT RETURN: -217.71%
- TOTAL INTRADAY RETURN: 480.26%
- OVERNIGHT DOMINANCE: -82.9% OF TOTAL RETURNS
- H_0 : AVERAGE OVERNIGHT RETURN = INTRADAY (NO DIFFERENCE).

WHY Z TEST?

1. CHI-SQUARE WORKS FOR CATEGORICAL DATA
2. T-TESTS ARE PERFECT WHEN YOU HAVE SMALL SAMPLES (UNDER 30-100 OBSERVATIONS)
3. ANOVA COMPARES MULTIPLE GROUPS SIMULTANEOUSLY WHERE X IS CATEGORICAL AND Y IS NUMBER

RESULT OF THE Z TEST



BASED ON THE Z TEST WHAT CAN WE GET?

BASED ON - **INTRADAY CAPTURE**

EXECUTION RULES:

- ENTRY: BUY AT 9:30 AM MARKET OPEN
- EXIT: SELL AT 4:00 PM MARKET CLOSE

BASED ON - **VOLUME PREDICTS VOLATILITY**

IF PRE-MARKET VOLUME > 20-DAY AVERAGE:

- REDUCE POSITION TO 50%
- WIDEN STOPS TO 2X NORMAL

ELSE:

- TRADE FULL POSITION
- NORMAL STOP LOSS AT -2%

EVERY TRADE SHOULD BE GROUNDED IN RESEARCH AND STRATEGY. "AN INVESTOR WITHOUT INVESTMENT OBJECTIVES IS LIKE A TRAVELER WITHOUT A DESTINATION." – RALPH SEGER.